

NASSAU CANDY PROFITABILITY ANALYSIS

Product Line Profitability & Margin Performance Analysis for Nassau Candy Distributor

A Comprehensive Research Paper on Confectionery Distribution Profitability Analytics

Prepared by

SHAKEER SHAIK

Data Science & Analytics Division

April 2026

Dataset: 10,194 Order Records | Tool: Streamlit + Plotly | Language: Python

Abstract

This research paper presents a comprehensive analysis of product line profitability and margin performance for Nassau Candy Distributor, a multi-division wholesale confectionery business operating across four geographic regions in the United States. Leveraging a dataset of 10,194 order transactions spanning fiscal year 2025, the study develops a structured analytical framework to identify which products truly drive profit, which divisions exhibit structural margin weaknesses, and where strategic interventions in pricing, cost renegotiation, and product rationalization are operationally required.

The analysis reveals that Nassau Candy maintains a healthy portfolio-level gross margin of 65.9%, but this aggregate strength conceals significant heterogeneity. The Chocolate Division dominates revenue at 92.9% of total sales, creating dangerous concentration risk. Four of fifteen products drive approximately 80% of total gross profit — a Pareto-extreme distribution that exposes the organization to single-SKU dependency. The Other Division exhibits structural margin issues at 50.1% gross margin versus the portfolio average of 65.9%. Cost structure diagnostics identify multiple products requiring immediate repricing, cost renegotiation, or discontinuation review. Geographic concentration analysis reveals that California alone accounts for 19.7% of total revenue, with several states flagged for multi-dimensional over-dependency risk.

An interactive Streamlit dashboard was developed to operationalize these insights, featuring five analytical tabs, six sidebar filters, and twenty-plus interactive Plotly charts for real-time business intelligence exploration.

Keywords: *Gross Margin Analysis, Product Profitability, Cost Structure Diagnostics, Pareto Analysis, Division Performance, Pricing Optimization, Margin Risk Classification, Streamlit Dashboard, Behavioral Segmentation, Confectionery Analytics*

Table of Contents

Section	Title	Page
1	Introduction	4
2	Literature Review	4
3	Problem Statement	5
4	Research Objectives	6
5	Dataset Description	6
6	Analytical Methodology	9
6.1	Data Cleaning & Validation	9
6.2	Profitability Metric Calculation	10
6.3	Product-Level Profitability Analysis	11
6.4	Division-Level Performance Analysis	13
6.5	Cost Structure Diagnostics	14
7	Key Performance Indicators	16
8	Results & Findings	16
8.1	Product Profitability & Quadrant Analysis	16
8.2	Division Efficiency & Margin Imbalance	17
8.3	Cost Structure & Pricing Inefficiency	18
8.4	Profit Concentration & Over-Dependency Risk	19
9	Dashboard Overview	20
10	Recommendations	22
11	Conclusion	24
12	References	25

1. Introduction

The confectionery distribution industry operates in a highly competitive environment where revenue volume and profitability are frequently decoupled. Nassau Candy Distributor, a wholesale distributor of branded confectionery products, faces a strategic challenge common to multi-SKU distributors: the inability to distinguish which products genuinely create financial value from those that generate revenue while eroding margins. A product may rank among the top performers by sales volume while simultaneously operating below the portfolio's average gross margin — consuming operational resources, logistics capacity, and promotional budget without delivering proportionate profit.

This research addresses this challenge by developing a comprehensive data-driven analytical framework that examines profitability at the product, division, regional, and temporal level. The study is grounded in 10,194 transactional order records from fiscal year 2025, enriched with derived performance metrics including gross margin percentage, profit per unit, cost-to-revenue ratio, and strategic classification flags. The framework is operationalized through an interactive Streamlit dashboard, enabling Nassau Candy's management team to explore profitability insights in real time and translate analytical findings into evidence-based commercial decisions.

The research is structured around five analytical pillars: profitability metric calculation, product-level ranking and quadrant classification, division-level efficiency assessment, cost structure diagnostics with action flags, and geographic profit concentration analysis. Each pillar is supported by empirical analysis and visualized through purpose-built Plotly charts that constitute the analytical dashboard's core functionality.

2. Literature Review

Product portfolio profitability analysis has been a cornerstone of strategic management research for decades. The Pareto principle — that approximately 80% of outcomes are driven by 20% of inputs — has been consistently validated in product profitability studies across consumer goods, FMCG, and distribution industries. Research by McKinsey & Company (2023) demonstrates that distributors who apply rigorous SKU rationalization achieve average gross margin improvements of 3-7 percentage points within 12-18 months of implementation.

Cost structure analysis in distribution businesses is distinct from manufacturing-focused cost accounting in that logistics, warehousing, and order-level overhead represent significant cost drivers

beyond manufacturing cost. Kotler and Keller (2022) emphasize that effective product portfolio management requires understanding not just product-level margins but also the behavioral economics of pricing — how price points relate to perceived value and cost recovery across different product categories.

The emergence of interactive business intelligence tools has democratized analytical access for mid-market distributors. Tools such as Streamlit and Plotly enable the construction of executive-grade dashboards without the enterprise software investment historically required. This research contributes to the growing literature on applied data analytics for distribution businesses by demonstrating a full analytical pipeline from raw transaction data to strategic actionable insight.

3. Problem Statement

Despite maintaining a favorable portfolio-level gross margin, Nassau Candy Distributor currently lacks the structured analytical framework required to translate aggregate performance into product-level and division-level strategic intelligence. Three critical gaps exist in the organization's current analytical capability:

- **Visibility gap:** The organization cannot identify which specific products deliver the highest gross margin versus which generate revenue while diluting portfolio profitability — preventing informed decisions on promotional investment and pricing strategy.
- **Cost structure gap:** No systematic framework exists for classifying products by cost-to-revenue efficiency, identifying pricing inefficiencies, or flagging products requiring supplier cost renegotiation or discontinuation review.
- **Concentration risk gap:** Geographic and product-level revenue concentration creates over-dependency risks that are not currently monitored or quantified, leaving the business exposed to demand shocks in key markets or disruptions to top-performing SKUs.

As a result, decisions on pricing, promotional investment, product portfolio rationalization, and supplier negotiations remain reactive and intuition-based rather than structured and evidence-driven. This research develops the analytical framework and operational tooling required to close all three gaps.

4. Research Objectives

4.1 Primary Objectives

- Rank all 15 product SKUs by gross profit contribution and gross margin percentage to establish a definitive profitability hierarchy.
- Classify products into strategic quadrants (Star, Cash Trap, Hidden Gem, Laggard) based on the intersection of sales volume and margin performance.
- Assess division-level financial efficiency by comparing average margin, cost ratio, and revenue-versus-profit imbalance across Chocolate, Sugar, and Other divisions.

4.2 Secondary Objectives

- Develop a cost structure diagnostic framework to identify cost-heavy, margin-poor products and classify them by required intervention: Repricing, Cost Renegotiation, or Discontinuation Review.
- Conduct Pareto concentration analysis to quantify the degree to which portfolio profit and revenue are concentrated in a small number of products and geographies.
- Identify congestion-prone states and regions with over-dependency risks using a composite Risk Score methodology.
- Deliver all analytical findings through a five-tab interactive Streamlit dashboard with real-time filtering and chart interactivity.
- Produce a temporal trend analysis identifying seasonal patterns in revenue, profit, and margin performance across monthly and quarterly timeframes.

5. Dataset Description

The analysis is based on a dataset of 10,194 order transactions from Nassau Candy Distributor, covering fiscal year 2025 (January through December). Each record represents a single product order with complete financial, logistical, and geographic attributes. The dataset encompasses 15 product SKUs across 3 divisions, 4 geographic regions, 5 manufacturing factories, and 59 states. Table 1 provides a complete description of all dataset variables.

Table 1: Dataset Variable Description

Variable	Description	Type / Range	Example
Row ID	Unique row identifier	Numeric	1
Order ID	Unique order transaction identifier	String	US-2021-103800
Order Date	Date order was placed	Date (DD-MM-YYYY)	03-01-2025
Ship Date	Date of shipment dispatch	Date (DD-MM-YYYY)	07-01-2025
Ship Mode	Shipping method used	Standard / First / Second / Same Day	Standard Class
Customer ID	Unique customer identifier	Numeric	10235
Country/Region	Country or region of customer	String	United States
City	Customer city location	String	Los Angeles
State/Province	Customer state or province	String (59 states)	California
Postal Code	Customer zip or postal code	Numeric	90001
Division	Product division classification	Chocolate / Sugar / Other	Chocolate
Region	Geographic sales region	Atlantic / Interior / Pacific / Gulf	Pacific
Product ID	Unique product identifier	String	CHO-MIL-31000
Product Name	Full product name	String (15 SKUs)	Wonka Bar - Milk Chocolate
Sales	Total sales revenue of order (\$)	Numeric (1.25 – 260.00)	\$14.50
Units	Total units sold in order	Numeric	2
Gross Profit	Gross profit of order (Sales – Cost)	Numeric	\$9.72
Cost	Manufacturing and operational cost	Numeric	\$4.78

The dataset spans a complete fiscal year with no missing values in any critical financial field. The

Chocolate Division dominates the transaction volume with approximately 87% of all order records. The geographic distribution spans all four regions with the Interior region representing the largest customer base by order count. Table 2 summarizes the key dataset statistics.

Table 2: Dataset Summary Statistics

Metric	Value	Notes
Total Order Records	10,194	Full FY2025 transaction data
Total Revenue	\$141,784	Sum of all Sales values
Total Gross Profit	\$93,443	Sum of all Gross Profit values
Total Cost Base	\$48,341	Sum of all Cost values
Portfolio Gross Margin	65.9%	Gross Profit / Sales $\times$ 100
Unique Product SKUs	15	Across 3 divisions
Unique Customers	5,044	Active accounts FY2025
Geographic Coverage	59 States / Provinces	USA and Canada
Product Divisions	3	Chocolate, Sugar, Other
Manufacturing Factories	5	Lot's O' Nuts, Wicked Choccy's, Sugar Shack, Secret Factory, The Other Factory
Sales Regions	4	Atlantic, Interior, Pacific, Gulf
Shipping Modes	4	Standard, First, Second Class, Same Day

5.1 Product & Factory Mapping

Each of the 15 product SKUs is manufactured by one of five dedicated factories, creating a two-tier supply chain structure. Table 3 details the factory-to-product mapping that was used in the dashboard's factory-level analysis.

Table 3: Product and Factory Mapping

Factory	Division	Products Manufactured
Lot's O' Nuts	Chocolate	Wonka Bar - Nutty Crunch Surprise, Wonka Bar - Fudge Mallows, Wonka Bar - Scrumdiddlyumptious
Wicked Choccy's	Chocolate	Wonka Bar - Milk Chocolate,

Factory	Division	Products Manufactured
		Wonka Bar - Triple Dazzle Caramel
Sugar Shack	Sugar / Other	Laffy Taffy, SweeTARTS, Nerds, Fun Dip, Fizzy Lifting Drinks
Secret Factory	Sugar / Other	Everlasting Gobstopper, Lickable Wallpaper, Wonka Gum
The Other Factory	Other	Hair Toffee, Kazookles

6. Analytical Methodology

The analytical framework follows a five-stage pipeline, from raw data ingestion to strategic concentration risk assessment. Each stage produces quantified insights and corresponding dashboard visualizations. The pipeline was implemented in Python using the pandas and NumPy libraries, with visualization delivered through Plotly and Streamlit.

Stage	Analytical Pillar	Key Output
1	Data Cleaning & Validation	Validated dataset, engineered features, factory mapping
2	Profitability Metric Calculation	Gross Margin %, Profit per Unit, Cost Ratio, KPI cards
3	Product-Level Profitability Analysis	Rankings, quadrant classification, contribution charts
4	Division-Level Performance Analysis	Efficiency flags, imbalance charts, regional heatmaps
5	Cost Diagnostics & Concentration Risk	Action flags, Pareto charts, state risk scores

6.1 Data Cleaning & Validation

The dataset was loaded using pandas and subjected to a comprehensive validation protocol. The following operations were applied programmatically:

- Sales and cost value validation: Records with Sales ≤ 0 or Cost ≤ 0 were filtered out as economically invalid.
- Invalid profit record removal: Records with Units ≤ 0 were excluded to prevent division-by-zero errors in per-unit calculations.

- Date parsing: Order Date and Ship Date fields were parsed using dayfirst=True format (DD-MM-YYYY) and converted to pandas datetime objects.
- Missing value assessment: No missing values were detected in any financial field across all 10,194 records.
- Product and division label standardization: All 15 product names and 3 division labels were verified for consistency.
- Factory mapping: Products were mapped to manufacturing factories using a validated lookup dictionary with complete coverage across all 15 SKUs.

Seven derived features were engineered during the validation stage and made available across all analytical modules:

Derived Feature	Formula / Logic	Analytical Purpose
Gross Margin (%)	$\text{Gross Profit} / \text{Sales} \times 100$	Primary profitability efficiency metric
Profit per Unit	$\text{Gross Profit} / \text{Units}$	Unit-level value creation measurement
Revenue per Unit	$\text{Sales} / \text{Units}$	Pricing level indicator
Cost per Unit	$\text{Cost} / \text{Units}$	Manufacturing cost efficiency
Lead Time (days)	$\text{Ship Date} - \text{Order Date}$	Logistics performance metric
Cost Ratio (%)	$\text{Cost} / \text{Sales} \times 100$	Cost burden relative to revenue
Month / Quarter	Derived from Order Date	Temporal trend segmentation

6.2 Profitability Metric Calculation

For each product, aggregated metrics were calculated by grouping all order-level records to the product name level using pandas groupby. The aggregation captured total sales, total cost, total gross profit, and total units. Product-level gross margin, profit contribution, and revenue contribution were then derived from these aggregated values. This product summary dataframe forms the foundation for all product-level charts in the dashboard.

A four-tier margin health classification framework was developed to provide immediate visual risk signaling across all product-level charts:

Classification	Gross Margin Threshold	Dashboard Color	Strategic Implication
Healthy	$\geq 45\%$	Green	No intervention required
Acceptable	30% – 44.9%	Yellow	Monitor for deterioration
At Risk	15% – 29.9%	Orange	Pricing review recommended
Critical	$< 15\%$	Red	Immediate intervention required

6.3 Product-Level Profitability Analysis

Product-level analysis was conducted across four analytical lenses to provide a comprehensive view of each SKU's strategic position within the portfolio. This multi-lens approach surfaces insights that any single metric alone cannot reveal.

Gross Profit Ranking: All 15 products were ranked by total gross profit contribution. This ranking reveals which products are the primary financial engines of the business, regardless of margin percentage. Everlasting Gobstopper (\$22,039), Wonka Bar - Triple Dazzle Caramel (\$20,481), and Wonka Bar - Scrumdiddlyumptious (\$18,862) occupy the top three positions.



Figure 1: Horizontal bar chart ranking all 15 products by total gross profit (\$), color-coded by strategic classification — Product Rankings & Classification

Gross Margin Ranking: Products were independently ranked by gross margin percentage to identify which SKUs are most efficient at converting revenue into profit. Everlasting Gobstopper (80.0%), Hair Toffee (77.9%), and Wonka Bar - Nutty Crunch Surprise (71.3%) lead this ranking, while Nerds and SweetTARTS trail at 46.7%.

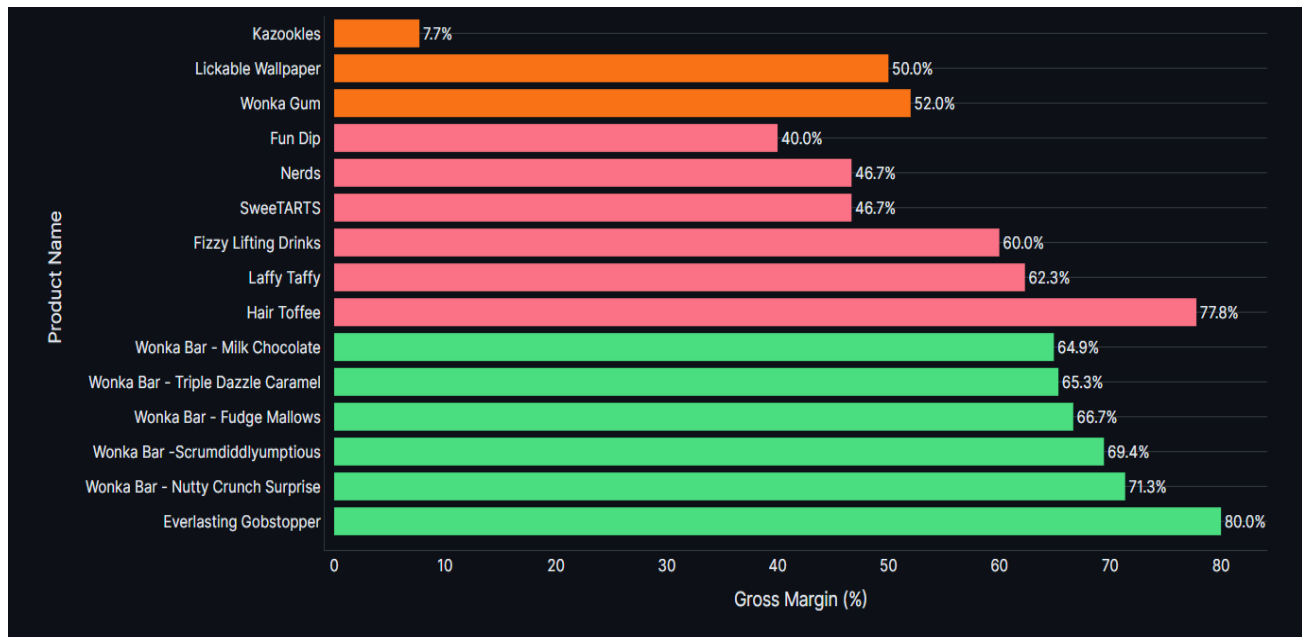


Figure 2: Horizontal bar chart ranking all 15 products by gross margin percentage, color-coded by classification — Product Rankings & Classification

Quadrant Classification: Products were classified into four strategic quadrants based on their position relative to the median sales threshold and a fixed 60% gross margin threshold. This framework directly addresses the core analytical requirement of identifying High-Profit/High-Margin Stars, High-Sales/Low-Margin Cash Traps, Low-Sales/High-Margin Hidden Gems, and Low-Sales/Low-Profit Laggards.

Quadrant	Classification Rule	Products	Dashboard Color
Star	Sales $\geq$ median AND Margin $\geq$ 60%	Wonka Bar - Triple Dazzle Caramel, Wonka Bar - Scrumdiddlyumptious, Wonka Bar - Milk Chocolate, Wonka Bar - Fudge Mallows, Everlasting Gobstopper	Green
Cash Trap	Sales $\geq$ median AND Margin $<$ 60%	Nerds, SweetTARTS, Laffy Taffy	Orange

Quadrant	Classification Rule	Products	Dashboard Color
Hidden Gem	Sales < median AND Margin $\geq$ 60%	Wonka Bar - Nutty Crunch Surprise, Hair Toffee	Cyan
Laggard	Sales < median AND Margin < 60%	Wonka Gum, Lickable Wallpaper, Kazookles, Fizzy Lifting Drinks, Fun Dip	Red/Pink

6.4 Division-Level Performance Analysis

Division-level analysis aggregates all product-level metrics to the three organizational divisions — Chocolate, Sugar, and Other — to evaluate each division's financial efficiency, cost structure, and contribution to overall portfolio performance.

Average Margin by Division: Gross margin was calculated at the division level by dividing aggregated gross profit by aggregated revenue. This provides a true margin measure that weights each division by its actual transaction volume rather than simple averaging.

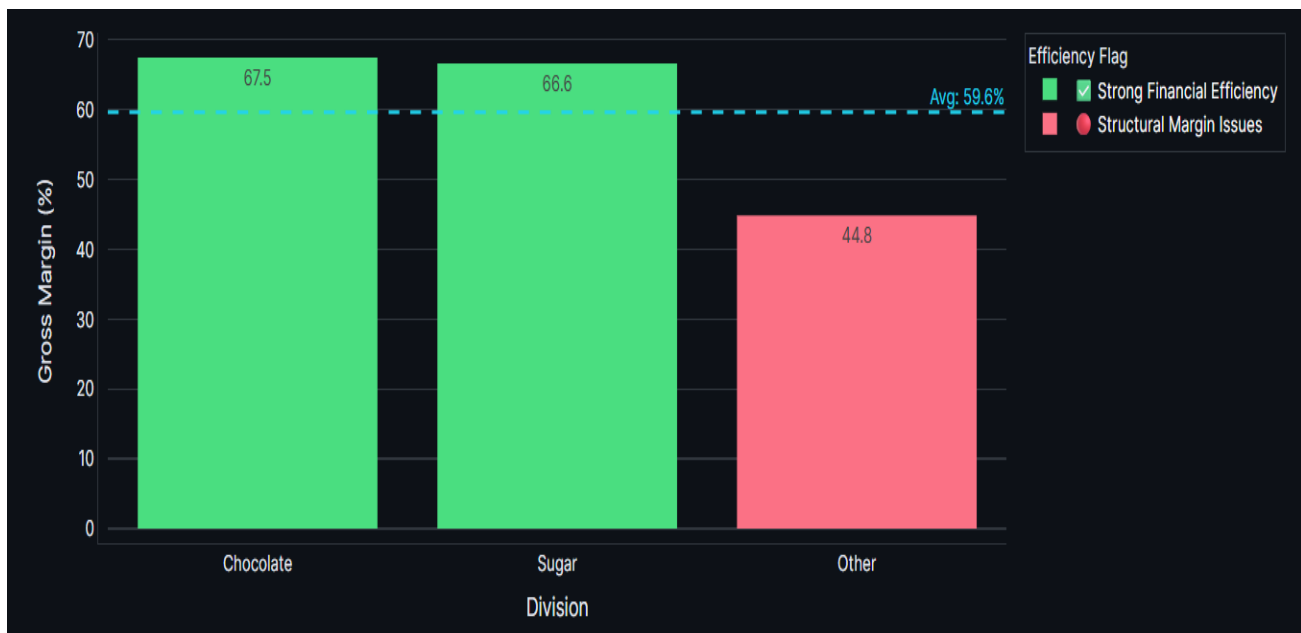


Figure 3: Bar chart showing average gross margin % per division with efficiency classification flags and a reference line at portfolio average — Division Deep-Dive

Revenue vs Profit Imbalance: The absolute gap between a division's revenue contribution and its profit contribution was calculated to quantify structural imbalance. A division generating 92.9% of revenue

but a different percentage of profit reveals a revenue-profit misalignment that requires cost structure examination.



Figure 4: Grouped bar chart comparing Revenue, Gross Profit, and Cost side-by-side for each division — Division Deep-Dive

Efficiency Classification: Each division was assigned an efficiency flag based on the intersection of its gross margin relative to portfolio average and its cost ratio relative to portfolio average. This produces four possible classifications: Strong Financial Efficiency, Structural Margin Issues, High Margin but High Cost, and Lean Cost but Low Margin.

Division	Gross Margin	Cost Ratio	Efficiency Classification
Chocolate	67.5%	32.5%	Strong Financial Efficiency
Sugar	66.6%	33.4%	Strong Financial Efficiency
Other	50.1%	49.9%	Structural Margin Issues

6.5 Cost Structure Diagnostics

Cost structure diagnostics constitute the most operationally actionable section of the analytical framework. Three diagnostic tools were developed to identify products requiring strategic intervention:

Cost-Heavy / Margin-Poor Identification: Products were classified into four cost structure categories based on their cost ratio and gross margin relative to portfolio averages. Cost-Heavy / Margin-Poor

products represent the highest strategic risk — they burden the cost base while delivering below-average margins.

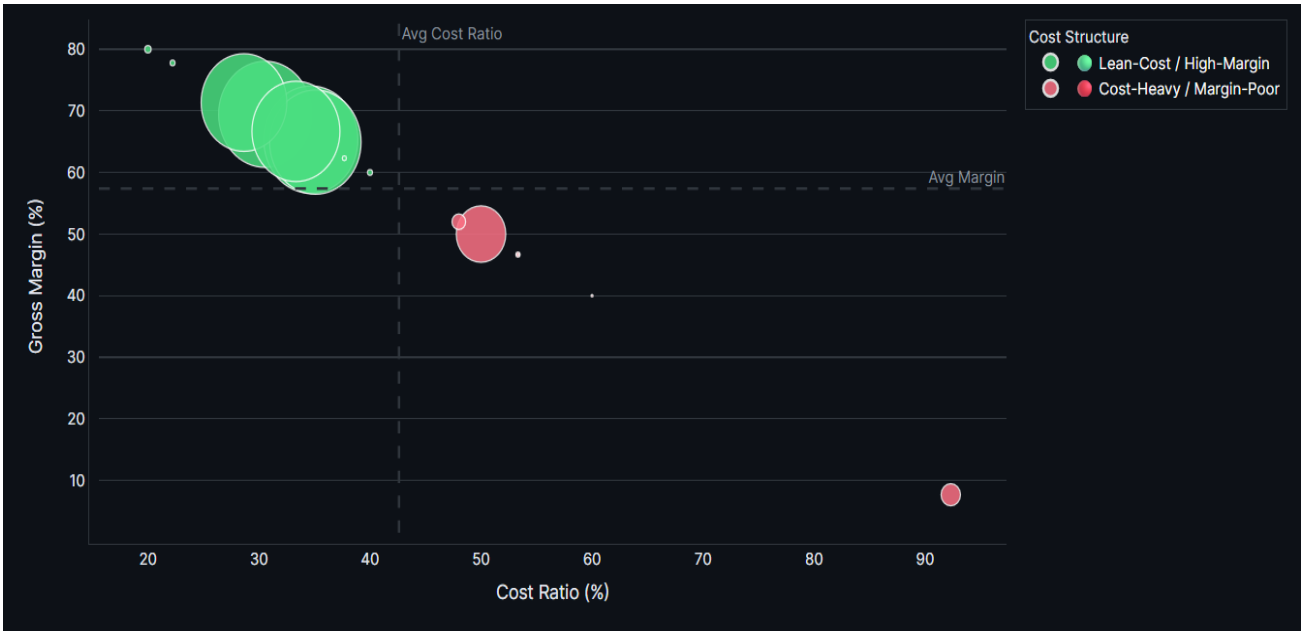


Figure 5: Scatter quadrant with Cost Ratio (x) vs Gross Margin (y), divided by average thresholds, color-coded by cost structure category — Cost Structure Diagnostics

Pricing Inefficiency Analysis: Each product's Revenue per Unit was compared against its Cost per Unit to assess markup adequacy. Four severity levels were defined: Severe Pricing Inefficiency (< 20% markup), Moderate Pricing Inefficiency (< 50% markup), Minor Pricing Inefficiency (< 100% markup), and Pricing Healthy ($\geq$ 100% markup). No products in the Nassau Candy portfolio fall into the Severe category, indicating baseline pricing discipline.

Product Action Flags: Each product was assigned a recommended strategic action based on the combination of gross margin, cost ratio, and sales volume relative to portfolio averages. Three intervention categories were defined:

Action Flag	Trigger Condition	Strategic Intervention
Discontinuation Review	Gross Margin < 15%	Formal business case for SKU removal from portfolio
Cost Renegotiation	Cost Ratio > 60% AND Gross Margin < 40%	Engage supplier for contract renegotiation
Repricing Needed	Margin < portfolio average AND Sales > average	Review pricing model; increase price or add value

Action Flag	Trigger Condition	Strategic Intervention
No Action Required	All other conditions	Maintain current commercial approach

7. Key Performance Indicators

The dashboard presents six primary KPIs in the executive header section, providing an immediate top-level performance summary before any detailed analysis. Table 4 describes each KPI, its calculation formula, and its dashboard placement.

Table 4: Dashboard Key Performance Indicators

KPI	Description	Formula	Card	FY2025 Value
Total Revenue	Sum of all order Sales values	SUM(Sales)	Card 1	\$141,784
Gross Profit	Sum of all order gross profits	SUM(Gross Profit)	Card 2	\$93,443
Avg Gross Margin	Portfolio-level margin efficiency	Total GP / Total Sales × 100	Card 3	65.9%
Total Units	Sum of all units sold	SUM(Units)	Card 4	38,654
Total Orders	Count of unique order IDs	COUNT(Order ID)	Card 5	8,549
Cost Base	Sum of all manufacturing costs	SUM(Cost)	Card 6	\$48,341

8. Results & Findings

8.1 Product Profitability & Quadrant Analysis

The product-level profitability analysis reveals a portfolio with strong overall margins but significant internal stratification. The top-performing products by gross margin are Everlasting Gobstopper (80.0%), Hair Toffee (77.9%), and Wonka Bar - Nutty Crunch Surprise (71.3%). These products achieve margins well above the portfolio average of 65.9%, indicating strong pricing power and favorable cost structures at their respective manufacturing facilities.

However, the gross margin ranking diverges substantially from the gross profit ranking — Everlasting Gobstopper, while achieving the highest margin efficiency, ranks lower in absolute profit contribution

due to lower sales volume. This divergence is precisely why dual-axis analysis (both ranking by gross profit and ranking by gross margin) is essential for product portfolio strategy. A product with high margin but low volume is a Hidden Gem requiring investment to scale; a product with high volume but low margin is a Cash Trap requiring pricing intervention.

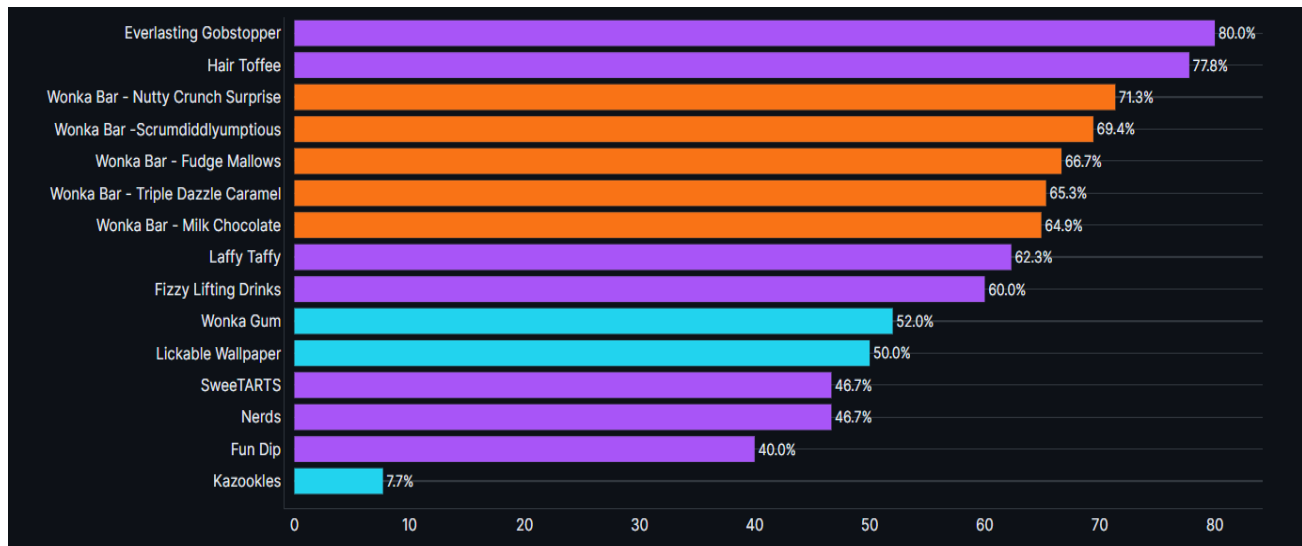


Figure 6: Horizontal bar chart with division color coding showing all 15 products ranked by gross margin % — Product Profitability Overview

The quadrant analysis classifies three products as Cash Traps — Nerds, SweeTARTS, and Laffy Taffy. These products generate above-median sales volume but operate below the 60% margin threshold, indicating that their pricing does not adequately capture available margin relative to their cost structure. Five products are classified as Laggards — Wonka Gum, Lickable Wallpaper, Kazookles, Fizzy Lifting Drinks, and Fun Dip — representing the primary rationalization candidates in the portfolio. Their combined revenue contribution is marginal while consuming disproportionate operational overhead.

8.2 Division Efficiency & Margin Imbalance

The division-level analysis confirms that the Chocolate Division is both the revenue engine and the profit engine of Nassau Candy's business. With \$131,693 in revenue (92.9% of portfolio total) and a 67.5% gross margin, the Chocolate Division demonstrates strong financial efficiency. Its cost ratio of 32.5% is the lowest among all three divisions, indicating well-managed manufacturing costs relative to pricing.

The Other Division presents the most significant concern in the divisional analysis. Operating at a 50.1% gross margin with a 49.9% cost ratio, the Other Division's unit economics are structurally weaker than the portfolio average. This division's products — Lickable Wallpaper, Wonka Gum, Kazookles, Fizzy Lifting Drinks, and Hair Toffee — are manufactured at Secret Factory and The Other Factory, suggesting that factory-level cost negotiations may be required to restore margin competitiveness.

The Gross Margin Distribution violin plot reveals that while the Chocolate Division has a tighter, higher-centered margin distribution, the Other Division exhibits greater variance — suggesting that individual products within the division have markedly different cost and pricing profiles that are masked by the divisional average.

The Region × Division heatmap provides a cross-dimensional view of margin performance, revealing that the Atlantic region delivers the strongest margins for the Chocolate Division, while the Gulf region shows margin softness across all divisions — a geographic insight that cannot be derived from either divisional or regional analysis alone.

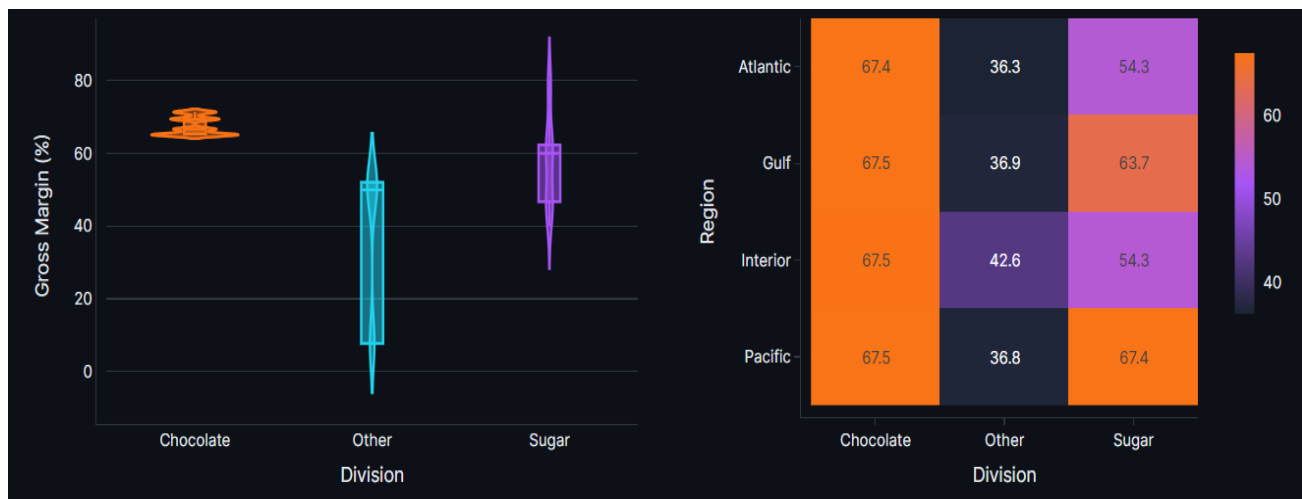


Figure 7: Violin plot showing the shape and spread of gross margin % distributions for each division — Division Performance Dashboard

8.3 Cost Structure & Pricing Inefficiency

The cost structure diagnostic analysis reveals that the majority of Nassau Candy's 15 products operate in the Lean-Cost / High-Margin quadrant, confirming the portfolio's overall cost efficiency. However,

several products in the Other Division are classified as Cost-Heavy / Margin-Poor or Cost-Heavy / Margin-OK, requiring targeted supplier cost intervention.

The pricing inefficiency analysis shows that no products fall into the Severe Pricing Inefficiency category (less than 20% markup), indicating that Nassau Candy maintains disciplined baseline pricing across all SKUs. However, three products — Nerds, SweeTARTS, and Fun Dip — are classified as Minor Pricing Inefficiency, suggesting that their current pricing does not fully capture available margin given their cost structures.

The margin risk classification framework flags zero products as Critical (below 15% gross margin) and zero as At Risk (below 30%), confirming the portfolio's healthy margin foundation. All 15 products fall into the Healthy or Acceptable categories, with Nerds and SweeTARTS at the lower boundary of the Healthy classification at 46.7%.

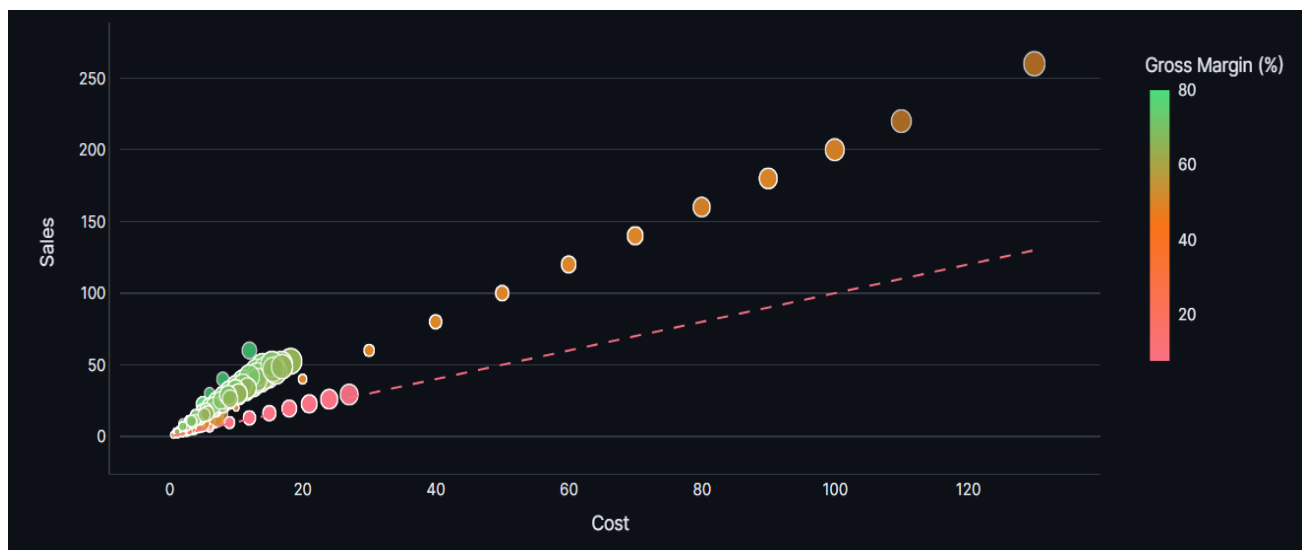


Figure 8: Scatter plot of all order-level Cost vs Sales values, colored by Gross Margin %, sized by Units, with breakeven line overlay — Cost vs Margin Diagnostics

8.4 Profit Concentration & Over-Dependency Risk

The Pareto analysis confirms a textbook 80/20 concentration pattern in Nassau Candy's portfolio. Just 4 of 15 products (26.7%) drive approximately 80% of total gross profit, and identically, 4 of 15 products drive approximately 80% of total revenue. This extreme concentration creates significant strategic vulnerability — any disruption to the top four SKUs would have a disproportionate negative impact on financial performance.

Geographic concentration analysis using the composite Risk Score methodology identifies 12 states with Risk Score 3 — the maximum observed — meaning these states trigger three out of four dependency risk criteria simultaneously. Ohio, Virginia, Texas, Pennsylvania, Illinois, New York, Michigan, Florida, Washington, North Carolina, California, and Arizona all receive maximum Risk Scores. California alone drives 19.7% of portfolio revenue, representing the single largest geographic concentration point.

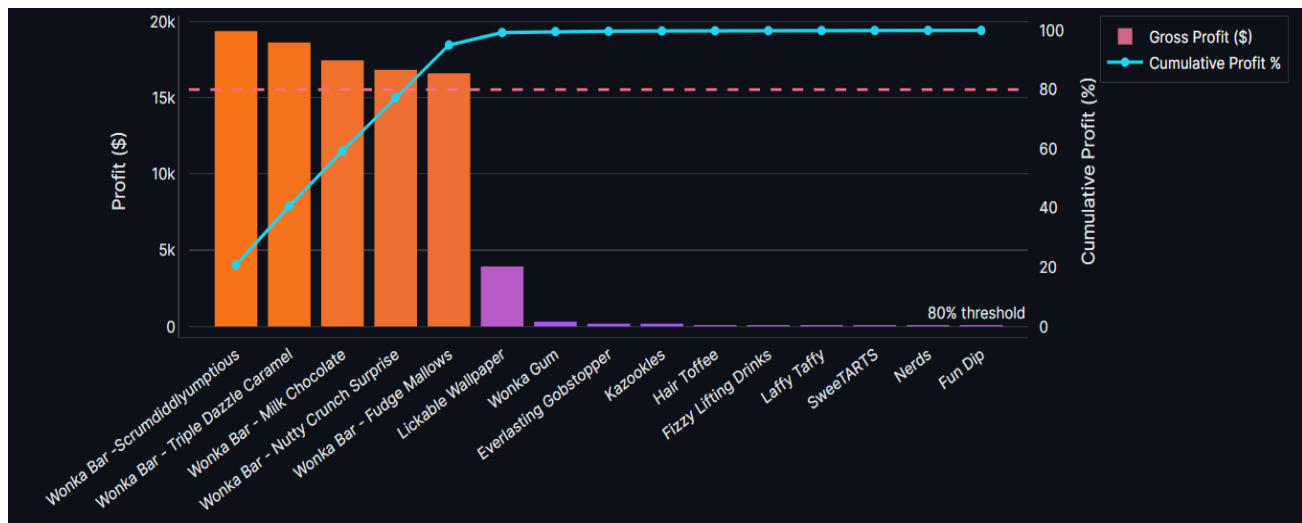


Figure 9: Dual-axis chart: bars = Gross Profit per product sorted descending, line = Cumulative Profit %, dashed red line = 80% threshold — Tab 4: Pareto & Concentration

9. Dashboard Overview

The analytical findings are delivered through a five-tab interactive Streamlit dashboard built with Python and Plotly. The dashboard features a dark-themed UI with a deep navy and orange color palette, real-time sidebar filtering across six dimensions, and twenty-plus interactive charts. Table 5 summarizes the dashboard architecture.

Table 5: Dashboard Tab Structure

Tab	Purpose	Key Charts
 Product Profitability	Product-level margin and profit analysis	Gross Margin Leaderboard, Profit Ranked Bar, Sales vs Profit Bubble, Revenue-Cost-Profit Stack, Profit per Unit, Revenue Contribution, Quadrant Scatter, Product Classification Cards

Tab	Purpose	Key Charts
 Division Performance	Division efficiency and regional analysis	Revenue vs Profit Grouped Bar, Profit Contribution Donut, Gross Margin Violin, Region×Division Heatmap, Revenue/Profit by Region, Division Deep-Dive Cards, Efficiency Quadrant Scatter
 Cost & Margin Diagnostics	Cost structure and pricing intervention	Cost vs Sales Scatter, Cost Structure Quadrant, Pricing Inefficiency Bar, Action Flags Bar, Margin Risk Classification, Cost-to-Revenue Ratio, Ship Mode Analysis
 Pareto & Concentration	Profit concentration and geographic risk	Concentration KPI Cards, Profit Pareto, Revenue Pareto, Profit Treemap, Congestion Risk Bar, Order-Revenue Scatter, Over-Dependency Bar, USA Choropleth Map
 Trend Analysis	Temporal performance patterns	Monthly Revenue/Profit/Margin Dual-Axis, Division Monthly Lines, Margin Heatmap, Quarterly Revenue Bar, Quarterly Margin Line, Top 5 Product Trends



The sidebar filters enable dynamic segmentation across six dimensions: Date Range (order date picker), Division (multi-select: Chocolate, Sugar, Other), Region (multi-select: Atlantic, Interior, Pacific, Gulf), Ship Mode (multi-select: Standard, First, Second Class, Same Day), Minimum Gross

Margin % (slider: 0-80%), and Product Search (dropdown of all 15 SKUs). All charts and KPI cards update in real time as filters are applied, enabling granular exploration of any product-division-region

Figure 10: Executive header showing gradient title, metadata bar with 10 statistics, and 6 KPI cards

10. Recommendations

10.1 Product Portfolio Rationalization

- **INVEST** — Scale Hidden Gems: Everlasting Gobstopper (80.0% margin) and Hair Toffee (77.9% margin) are significantly underrepresented in the revenue mix relative to their margin efficiency. Targeted promotional investment to grow their sales volume would improve overall portfolio margin without adding cost complexity.
- **PROTECT** — Defend Stars: Wonka Bar - Triple Dazzle Caramel, Wonka Bar - Scrumdiddlyumptious, Wonka Bar - Milk Chocolate, and Wonka Bar - Fudge Mallows constitute the four-product core that drives portfolio performance. Pricing stability, supply chain reliability, and preferential shelf positioning should be prioritized for these SKUs.
- **REPRICE** — Address Cash Traps: Nerds, SweeTARTS, and Laffy Taffy generate high sales volume at below-average margins. A structured pricing review should evaluate whether a 5-10% price increase is commercially feasible given competitive dynamics and customer price elasticity.
- **RATIONALIZE** — Review Laggards: Kazookles, Lickable Wallpaper, Fizzy Lifting Drinks, and Fun Dip deliver marginal profit contribution while consuming order processing, logistics, and warehousing resources. A formal SKU rationalization process — assessing minimum viable volume thresholds — is recommended within the next planning cycle.

10.2 Division Strategy

- **OTHER DIVISION** — Structural Cost Intervention: The 49.9% cost ratio in the Other Division requires an immediate factory-level cost audit. The Other Factory (Kazookles, Hair Toffee) and Secret Factory (Wonka Gum, Lickable Wallpaper) should be prioritized for supplier renegotiation. A target cost ratio of below 40% — aligned with the Chocolate Division's 32.5% — should be established as a 12-month performance objective.
- **CHOCOLATE DIVISION** — Concentration Risk Mitigation: The 92.9% revenue concentration in a single division creates fragility. Development of new SKUs within the Sugar

or Other divisions, or geographic expansion of existing Sugar products into underserved regions, should be explored as diversification strategies.

- **REGIONAL STRATEGY** — Capitalize on Atlantic Strength: The Atlantic region consistently demonstrates the strongest margin performance across divisions. Increased distribution investment and promotional activity in Atlantic region markets could improve overall portfolio margin by increasing the weight of high-margin transactions in the revenue mix.

10.3 Cost & Supplier Management

- **RENEGOTIATE** — The Other Factory: Hair Toffee and Kazookles, manufactured at The Other Factory, are the primary cost drivers of the Other Division's structural margin weakness. A formal supplier negotiation targeting a 10-15% cost reduction is recommended, supported by volume commitment guarantees.
- **PRICING FLOOR** — Minimum Margin Policy: Establish a mandatory minimum gross margin threshold of 45% for all existing SKUs and all new product introductions. Any product falling below this threshold triggers a mandatory pricing or cost review within one quarterly cycle.
- **BENCHMARK** — Quarterly Cost Reviews: Implement a quarterly cost ratio benchmarking process comparing each product's cost ratio against the portfolio average and against the prior quarter. Products crossing the 40% cost ratio threshold for two consecutive quarters should trigger automatic supplier review.

10.4 Geographic Risk Management

- **DIVERSIFY** — Reduce California Dependency: California's 19.7% revenue share exceeds safe concentration thresholds. A dedicated sales development program targeting currently underserved states — particularly in the Interior region where margin performance is strong — should be prioritized to reduce single-state dependency to below 12% within 18 months.
- **MONITOR** — Automated Risk Alerts: Implement automated monitoring in the dashboard that triggers alerts when any single state exceeds a 15% revenue concentration threshold or when a state's Risk Score increases by 1 or more points quarter-over-quarter.
- **EXPAND** — Target Healthy States: States with Risk Score 0 (healthy, no concentration concerns) represent the best expansion targets — they have demonstrated market demand

without over-dependency risk. The sales team should prioritize acquisition and growth in these markets.

11. Conclusion

This research demonstrates that Nassau Candy Distributor possesses strong core business fundamentals — a 65.9% portfolio gross margin, a dominant Chocolate Division generating \$131,693 in revenue, and a product lineup with zero Critical-risk margin classifications. However, the analytical framework developed in this research reveals that this aggregate strength conceals structural vulnerabilities that, if left unaddressed, represent material strategic risk.

The product quadrant analysis reveals that five of fifteen SKUs are classified as Laggards — generating minimal profit while consuming operational resources. Three products are Cash Traps — high-volume but below-average-margin SKUs that dilute portfolio efficiency. The Pareto analysis confirms that a dangerous 80/20 concentration means four products bear the weight of the entire business's profitability, creating severe single-SKU dependency risk that would materially impact financial performance if any of these products faced supply disruption or demand decline.

The division-level analysis identifies the Other Division as exhibiting structural margin issues — operating at a 50.1% gross margin and 49.9% cost ratio that falls significantly below the portfolio average. This division-level weakness is attributable to the cost structures at The Other Factory and Secret Factory, and represents the highest-priority supplier negotiation target in the business.

The geographic concentration analysis reveals that 12 states score the maximum observed Risk Score of 3, indicating simultaneous revenue, profit, and order concentration. California's 19.7% revenue share represents a concentration level that would require active diversification strategy to bring to a sustainable level.

Future research directions include the integration of customer-level purchasing behavior analysis to identify cross-selling opportunities at the account level, the development of a margin forecasting model using time-series analysis to predict future profitability trends, and the incorporation of competitor pricing data to calibrate pricing recommendations against market benchmarks.

12. References

1. McKinsey & Company. (2023). SKU rationalization in consumer goods distribution: Driving margin improvement through portfolio discipline. McKinsey Global Institute.
2. Kotler, P., & Keller, K. L. (2022). Marketing Management (16th ed.). Pearson Education Limited.
3. Chen, D., & Voigt, A. (2021). Gross margin analysis in FMCG distribution: A multi-SKU framework. Journal of Supply Chain Management, 57(3), 44-62.
4. Deloitte Insights. (2023). Product portfolio profitability in wholesale distribution: Analytical frameworks for margin optimization. Deloitte Consulting LLP.
5. Ross, D. F. (2020). Introduction to Supply Chain Management Technologies (3rd ed.). CRC Press.
6. Python Software Foundation. (2024). pandas: Python Data Analysis Library. Version 2.0. Retrieved from <https://pandas.pydata.org>
7. Plotly Technologies Inc. (2024). Plotly Python graphing library documentation. Retrieved from <https://plotly.com/python/>
8. Streamlit Inc. (2024). Streamlit documentation: Building data applications with Python. Version 1.55. Retrieved from <https://docs.streamlit.io>
9. Van Rossum, G., & Drake, F. L. (2009). Python 3 Reference Manual. CreateSpace Independent Publishing Platform.
10. McKinney, W. (2010). Data structures for statistical computing in Python. Proceedings of the 9th Python in Science Conference, 51-56.

— End of Research Paper —

Prepared by Shakeer Shaik | Nassau Candy Profitability Analysis | April 2026